

QMEA (Sampling distribution)

1. What is a sampling distribution?

- A. The distribution of all observations in a population
- B. The distribution of observations in one sample
- C. The probability distribution of a statistic obtained from all possible samples of a given size
- D. The distribution of population parameters

2. Why does the sample mean vary from one random sample to another?

- A. Because the population mean changes
- B. Because different samples contain different observations
- C. Because the population size changes
- D. Because the sample mean is always biased

3. What is the standard error of the mean?

- A. The standard deviation of the population
- B. The standard deviation of one sample
- C. The standard deviation of the sampling distribution of the sample mean
- D. The difference between the sample mean and population mean

4. What happens to the standard error of the sample mean when the sample size increases?

- A. It increases
- B. It decreases
- C. It remains unchanged
- D. It becomes equal to the population mean

5. What does the Central Limit Theorem (CLT) state?

- A. Every population is normally distributed
- B. Every sample is normally distributed
- C. The sampling distribution of the sample mean tends toward a normal distribution as sample size increases
- D. The population mean approaches zero as sample size increases

6. What does it mean to say that the sample mean is an unbiased estimator of the population mean?

- A. Every sample mean equals the population mean
- B. The sample mean can never be incorrect
- C. The average of all possible sample means equals the population mean
- D. Large samples always have exactly the population mean

7. A researcher repeatedly takes random samples of 30 students from a university and calculates the average GPA for each sample. The distribution of these averages is called:

- A. Population distribution
- B. Sample distribution
- C. Sampling distribution of the sample mean
- D. Normal distribution

8. A hospital records the blood pressure of every patient admitted during one year and examines the distribution of these individual blood pressure values. This is:

- A. Population distribution
- B. Sampling distribution
- C. Sampling distribution of the mean
- D. Standard error distribution

9. A researcher randomly selects 50 employees and plots the individual monthly salaries of those 50 employees. This graph represents:

- A. Population distribution
- B. Sample distribution
- C. Sampling distribution of the sample mean
- D. Distribution of population means

10. Two researchers repeatedly sample from the same population. Researcher A uses samples of $n = 10$, while Researcher B uses samples of $n = 100$. Which statement is most likely correct?

- A. Researcher A's sample means will have less variability
- B. Researcher B's sample means will have less variability
- C. Both sampling distributions must have the same variability
- D. Researcher B's population has a smaller standard deviation